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Assessing Critical Facility Accessibility and Road Network Criticality Under Flood-Induced Failures: A Resilience-Based Framework for Climate Change Adaptation

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Critical infrastructure, including transportation, energy supply, telecommunications, water supply, and government and emergency services, is essential for sustaining societal functioning and the well-being of people. Ensuring accessibility to critical facilities, such as health centers and fire stations, is particularly crucial for supporting life-saving and life-sustaining activities during and after disasters.

Flooding, a frequent and costly natural hazard, presents significant challenges to infrastructure accessibility. With climate change, the frequency and intensity of coastal and fluvial flooding are projected to increase, highlighting the need for a deeper understanding of its impacts on critical facilities. Ensuring these facilities remain accessible during and after flooding protects vulnerable populations and facilitates life-saving activities.

This study examines the impact of flood-induced disruptions on accessibility to health centers in Ennis, Ireland, under three scenarios: the Present Day, Mid-Range Future Scenario (MRFS), and High-End Future Scenario (HEFS). These scenarios reflect the anticipated increases in flood frequency and intensity for both coastal and fluvial flooding under future climate conditions. High-resolution flood maps are used to simulate the spatial extent of flooding and its effects on the road network.

A comprehensive framework is developed to assess accessibility loss and road criticality, integrating both physical and social vulnerabilities. This framework is designed to monitor the deterioration of territorial accessibility to critical infrastructure as a result of the cumulative elimination of road sections due to flooding. It incorporates a betweenness centrality (BC) metric to identify essential road segments that connect communities to critical services, helping to pinpoint areas most vulnerable to disruption. This approach enables the identification of key routes that are crucial for maintaining access to critical services during and after flooding events, enhancing preparedness and resilience. Social vulnerability is evaluated through a Social Vulnerability Index, emphasizing the disproportionate impacts on vulnerable populations, such as

the elderly, children, low-income households, the disabled, and those with bad and very bad health conditions.

The results reveal significant reductions in accessibility across all scenarios, with disparities worsening under future climate conditions. In the MRFS, the frequency and extent of accessibility disruptions increase compared to the present day, with travel times to health centers rising significantly, reflecting moderate climate impacts. In the HEFS, the situation becomes more dire, with a large portion of critical roads becoming impassable, and travel times to health centers and fire stations increasing substantially in the worst-affected areas.

These findings highlight the urgent need to improve infrastructure and implement proactive planning to address access challenges caused by flooding. Recommendations include upgrading critical roads, establishing real-time flood response systems, and temporarily relocating services during extreme flood events. By integrating social vulnerability into planning, this research offers practical guidance for fostering equitable community resilience and ensuring uninterrupted access to essential services during future climate-related disruptions. Emphasizing a resilience-based approach, the study provides actionable insights for policymakers and stakeholders in Ennis and similar urban areas to develop sustainable solutions that address both the physical impacts of flooding and the associated social vulnerabilities, underscoring the critical role of climate change adaptation strategies in safeguarding critical infrastructure and protecting vulnerable populations.